

University of Gujrat

Department of Software Engineering

PhiUSIIL Phishing URL Dataset Mapped to the CIA Triad

Submitted to:

Miss Ayesha Bibi

Submitted by:

Muhammad Aliyan

25021598-245

Section:

AB



Introduction to Dataset **3**

Dataset Introduction 3

Contents 3

Methodology and Mapping to CIA Triad **4**

Methodology 4

Findings Mapped to the CIA Triad 4

Confidentiality 4

Integrity 4

Availability 4

Conclusion **5**

Introduction to Dataset

Dataset Introduction

Dataset Name: PhiUSIIL Phishing URL Dataset

Source: UCI Machine Learning Repository and Kaggle, published by Arvind Prasad.

Contents

The dataset consists of 235,795 rows. Out of all these 134,850 (57.2%) are URLs and 100,945 (42.8%) are phishing URLs. Instead of directly analyzing network traffic, the dataset contains 54 characteristics commonly found in these URL strings and webpage code. This data is then used to train AI model to make it differentiate between actual and phishing URLs . Key columns include URLLength, IsDomainIP, URLSimiliarityIndex, and Label.

COLUMN NAME	EXPLANATION
URL, DOMAIN, AND TLD	These columns contains the raw data of the URL being analyzed like its URL, Domain and the TLD (Top Level Domain) consists of Top level domain like .com or .org etc.
URLLENGTH AND DOMAINLENGTH	These columns consists of the raw length of the domains and URL for these links. URL length is a pattern that helps distinguish between real and phishing websites
ISDOMAINIP	This domain consists of 0s and 1s where 0 represent that the URL's domain is not an IP and 1 represents a URL whose domain is an IP. Legit sites rarely use IP as their domains.
URLSIMILARITYINDEX	This assigns a score to the domain out of 100 based on how similar it is to a well-established website. Phishing websites use these similarly named domains to fool people.
LABEL (TARGET CLASS)	This column also consists of 0s and 1s where 0 is assigned to the phishing URL and 1 is assigned to the legitimate URL.

Methodology and Mapping to CIA Triad

Methodology

I took a look at the 54 columns of raw data to see exactly what I was dealing with. I used AI to link the technical details of the URLs to actual cyber threats and understand how it maps to the CIA triad. The main goal of the analysis is to show exactly how these phishing links compromise basic security when they hit a live network.

Findings Mapped to the CIA Triad

The dataset sorts links into two categories: safe or phishing. This setup lets us understand the exact damage caused if a user actually clicks on any of the 100,945 malicious links.

Confidentiality

The purpose of the 100,945 phishing links in this dataset is to break confidentiality. At its core, phishing works by tricking people into giving their passwords or personal data. Suspicious patterns in the data like hiding a website name behind an IP address (IsDomainIP) or using very long links (URLLength) are simply tactics designed to fool the user and and steal their information.

Integrity

Once hackers steal your passwords, they get the access to your data which allows them to steal your assets, mess with your data, or send fake message pretending as you. This would absolutely destory the integrity of the system. Columns like URLSimilairityIndex point out such URLs that have unusually similar domain as the poular domains. In this way, it helps avoid the attack on integrity.

Availability

Phishing is the main way hackers take systems completely offline. If just one of those 100,945 malicious links drops ransomware, it locks up network drives and blocks real users from accessing their files. That’s exactly why we have major downtimes on the systems we see in major cyberattacks.

Conclusion

The PhiUSIIL data proves that phishing isn't random. It relies on specific, recognizable patterns and makes up 42.8% of the analyzed links. The dataset exposes exactly how hackers slightly change the links and code to trick people. In the real world, missing these patterns always starts with data getting stolen (a breach of confidentiality). From there, it's an open door for hackers to destroy your files (integrity) and take everything offline (availability).